

true

Logical 1 (true)

Syntax

```
true
T = true(n)
T = true(sz)
T = true(sz1,...,szN)
T = true(__,like=p)
```

Description

true is shorthand for the logical value 1. [example](#)

T = true(**n**) is an n-by-n matrix of logical ones. [example](#)

T = true(**sz**) is an array of logical ones where the size vector, **sz**, defines size(**T**). For example, true([2 3]) returns a 2-by-3 array of logical ones. [example](#)

T = true(**sz1**,...,**szN**) is a sz1-by-...-by-szN array of logical ones where sz1,...,szN indicates the size of each dimension. For example, true(2,3) returns a 2-by-3 array of logical ones. [example](#)

T = true(__,like=**p**) returns an array of logical ones of the same sparsity as the logical variable **p** using any of the previous size syntaxes. [example](#)

Examples

[collapse all](#)

Generate Square Matrix of Logical Ones

Use true to generate a 3-by-3 square matrix of logical ones.

[Open in MATLAB Online](#)[Copy Command](#)

```
A = true(3)
```

[Get](#) ▼

A = 3×3 logical array

```
1  1  1
1  1  1
1  1  1
```

```
class(A)
```

[Get](#) ▼

```
ans =  
'logical'  
The result is of class logical.
```

Generate Array of Logical Ones with Arbitrary Dimensions

Use true to generate a 3-by-2-by-2 matrix of logical ones.

[Open in MATLAB Online](#)[Copy Command](#)

```
true(3,2,2)
```

[Get](#) ▼

```
ans = 3x2x2 logical array  
ans(:,:,1) =
```

```
1 1  
1 1  
1 1
```

```
ans(:,:,2) =
```

```
1 1  
1 1  
1 1
```

Alternatively, you can use a size vector to specify the size of the matrix.

```
true([3,2,2])
```

[Get](#) ▼

```
ans = 3x2x2 logical array  
ans(:,:,1) =
```

```
1 1  
1 1  
1 1
```

```
ans(:,:,2) =
```

```
1 1  
1 1  
1 1
```

Note that specifying multiple vector inputs returns an error.

Execute Logic Statement

true along with false can be used to execute logic statements.

Test the logical statement

$\sim(A \text{ and } B) = (\sim A) \text{ or } (\sim B)$

for A = true and B = false.

[Open in MATLAB Online](#)
[Copy Command](#)

```
~(true & false) == (~true) | (~false)
```

[Get](#)

```
ans = logical
      1
```

The result is logical 1 (true), since the logical statements on both sides of the equation are equivalent. This logical statement is an instance of De Morgan's Law.

Generate Logical Array of Selected Sparsity

Generate a logical array of the same sparsity as the selected array.

[Open in MATLAB Online](#)
[Copy Command](#)

```
A = logical(sparse(5,3));
whos A
```

[Get](#)

Name	Size	Bytes	Class	Attributes
A	5x3	41	logical	sparse

```
T = true(4,like=A);
whos T
```

[Get](#)

Name	Size	Bytes	Class	Attributes
T	4x4	184	logical	sparse

The output array T has the same sparse attribute and data-type as the specified array A.

Input Arguments

[collapse all](#)

n — Size of square matrix
integer

Size of square matrix, specified as an integer. n sets the output array size to n-by-n. For example, true(3) returns a 3-by-3 array of logical ones.

- If n is 0, then T is an empty matrix.
- If n is negative, then it is treated as 0.

Data Types: int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

sz — Size vector
row vector of integers

Size vector, specified as a row vector of integers. For example, `true([2 3])` returns a 2-by-3 array of logical ones.

- If the size of any dimension is 0, then **T** is an empty array.
- If the size of any dimension is negative, then it is treated as 0.
- If any trailing dimensions greater than 2 have a size of 1, then the output, **T**, does not include those dimensions.

Data Types: int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

sz1, ..., szN — Size inputs
comma-separated list of integers

Size inputs, specified by a comma-separated list of integers. For example, `true(2,3)` returns a 2-by-3 array of logical ones.

- If the size of any dimension is 0, then **T** is an empty array.
- If the size of any dimension is negative, then it is treated as 0.
- If any trailing dimensions greater than 2 have a size of 1, then the output, **T**, does not include those dimensions.

Data Types: int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

p — Prototype
logical variable

Prototype, specified as a logical variable.

Data Types: single | double | int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

Complex Number Support: Yes

Output Arguments

[collapse all](#)

T — Output of logical ones
scalar | vector | matrix | N-D array

Output of logical ones, returned as a scalar, vector, matrix, or N-D array.

Data Types: logical

Tips

- `true(n)` is much faster and more memory efficient than `logical(true(n))`.

Extended Capabilities

[expand all](#)

C/C++ Code Generation

Generate C and C++ code using MATLAB® Coder™.

GPU Code Generation

Generate CUDA® code for NVIDIA® GPUs using GPU Coder™.

Thread-Based Environment

Run code in the background using MATLAB® backgroundPool or accelerate code with Parallel Computing Toolbox™ ThreadPool.

GPU Arrays

Accelerate code by running on a graphics processing unit (GPU) using Parallel Computing Toolbox™.

Distributed Arrays

Partition large arrays across the combined memory of your cluster using Parallel Computing Toolbox™.

Version History

Introduced before R2006a

See Also

[false](#) | [logical](#)

Topics

[Class Support for Array-Creation Functions](#)

YOU MIGHT ALSO BE INTERESTED IN

[Converting Deep Learning Models Between PyTorch, TensorFlow, and MATLAB](#)

Import and export AI models between PyTorch®, TensorFlow™, and MATLAB®.

[Learn more](#)